

Module 4 – Big Ideas

Databases and SQL for Data Science

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The Big Picture

Module 4 is about one core shift:

Data scientists do not just analyze data. They read from and write to databases.

Up until now, most of our work has lived in:

- CSV files
- DataFrames
- Jupyter notebooks

In the real world, data lives in **relational databases**.

This module builds the bridge between:

- SQL (querying structured data)
 - Python (analysis and visualization)
 - Real-world database systems
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Big Idea 1 – Relational Thinking

A relational database is not just a file.

It is:

- Tables
- Columns with data types
- Rows of structured records
- Primary keys and foreign keys
- Relationships between entities

You should walk away understanding:

- What a relational model is
- How entities map to tables
- How attributes map to columns
- Why keys matter

This is the mental model behind everything else in this module.

Big Idea 2 – SQL Is the Language of Structured Data

SQL gives you structured control over structured data.

There are two main categories:

DDL (Data Definition Language)

- CREATE
- ALTER
- DROP
- TRUNCATE

Defines structure.

DML (Data Manipulation Language)

- INSERT
- SELECT
- UPDATE
- DELETE

Manipulates data.

If you understand this separation, SQL becomes much easier to reason about.

Big Idea 3 – Querying Is Set-Based Thinking

SQL is not procedural.

You are not looping.

You are not iterating.

You are declaring:

“Give me the subset of data that matches this logic.”

Core tools you must be comfortable with:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- COUNT, DISTINCT
- Built-in functions (SUM, AVG, MIN, MAX, ROUND, etc.)
- Subqueries / joins

SQL is about describing the result you want — not writing step-by-step instructions.

Big Idea 4 – Multi-Table Reasoning

Real databases rarely have one table.

You must understand:

- How to combine tables
- Implicit joins
- JOIN operators
- Table aliases
- Column qualification
- Subqueries across tables

This is where relational databases become powerful.

If you can reason across multiple tables, you can answer real business questions.

Big Idea 5 – SQL + Python Integration

This is where this course goes beyond a typical SQL class.

You are not learning SQL in isolation.

You are learning:

1. SQL Magic in Jupyter

- `%sql`

- `%%sql`

- Passing Python variables into SQL
 - Converting results to DataFrames
2. DB-API (`sqlite3`)
- Connecting programmatically
 - Executing queries
 - Committing transactions
 - Closing connections
3. Pandas integration
- `read_sql`
 - EDA on query results
 - Visualization

This is how real workflows operate.

SQL stores and filters.

Python analyzes and visualizes.

Big Idea 6 – Real-World Data Is Messy

You will encounter:

- Mixed-case column names
- Spaces in column names
- Metadata tables
- CSV imports
- Data type issues

- Quoting and escaping

This is intentional.

You need to get comfortable operating in imperfect schemas.

The goal is not textbook SQL.

The goal is operational SQL.

Big Idea 7 – The Final Project Is Workflow Mastery

The final project tests integration.

You must:

- Create a SQLite database
- Load three datasets
- Create tables
- Write correct queries
- Answer structured questions

This is not about memorizing syntax.

It is about demonstrating:

- Schema understanding
- Query correctness
- Multi-table reasoning
- SQL + Python integration

What You Should Be Comfortable With By the End

You should be able to:

- Design a simple relational table
- Load CSV data into SQLite
- Write SELECT queries with filtering and aggregation
- Use GROUP BY correctly
- Write subqueries
- Combine tables
- Connect Python to a database
- Pull results into Pandas
- Visualize query results

If you can do these things, you have a functional database toolkit.

The Core Mindset

Do not try to memorize every SQL clause.

Instead, aim to understand:

- How relational data is structured
- How to think in sets
- How to describe the result you want
- How to integrate SQL into a Python workflow

This module is about building a database foundation that you will reuse in:

- Data engineering

- Backend systems
- ML pipelines
- Analytics platforms

This is not just SQL.

This is database literacy.